

# ĆWICZENIA 3

①

Zad 1. Wprowadzając współrzędne biegunowe, obliczyć całki

1)  $\iint_D e^{-x^2-y^2} dx dy$ ,  $D = \{(x,y): 3 \leq x^2+y^2 \leq 4\}$ .

2)  $\iint_D \sqrt{x^2+y^2} dx dy$ ,  $D = \{(x,y): x^2+y^2 \leq 2y, x \geq 0\}$ .

3)  $\iint_D xy^2 dx dy$ ,  $D = \{(x,y): x^2+y^2 \leq 4, x \geq 0, y \geq 0\}$ .

4)  $\iint_D (x^2+y^2) dx dy$ ,  $D = \{(x,y): y \leq x^2+y^2 \leq x, y \geq 0\}$ .

wsk:  $a^4 - b^4 = (a^2 + b^2)(a^2 - b^2)$

$\sin^2 \alpha + \cos^2 \alpha = 1$ ,  $\cos^2 \alpha - \sin^2 \alpha = \cos 2\alpha$

Zad 2. Obliczyć pole obszaru D

1)  $D = \{(x,y): x+y \leq 3, y^2 \leq 4x, y \geq 0\}$ .

2) D - obszar ograniczony krzywymi:  $y=e^x$ ,  $y=\ln x$ ,  $x+y=1$ ,  $x=2$ .

3)  $D = \{(x,y): x^2+y^2 \leq 2x, y \geq x\}$ .

Zad 3. Obliczyć objętość bryły  $E \subset \mathbb{R}^3$ , gdzie

~~E~~ E jest bryłą ograniczoną poniższymi:

1)  $x^2+y^2=1$ ,  $x+y+z=3$ ,  $z=0$ .

2)  $x^2+y^2=2y$ ,  $z=x^2+y^2$ ,  $z=0$ .

3)  $z=e^{-x^2-y^2}$ ,  $x^2+y^2=9$ ,  $z=-1$ .

4)  $x+y+z=1$ ,  $x=0$ ,  $y=0$ ,  $z=0$ ,  $x+y=1$ .

5)  $x^2+y^2=4x$ ,  $\sqrt{x^2+y^2}=3z$ ,  $z=-2$ .

Zad 4. Obliczyć masę i współrzędne środka ciężkości  
jednorodnego obszaru  $D$ , gdzie:

②

1)  $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq e^x\}$ .

2)  $D = \{(x, y) : x^2 \leq y \leq 1\}$ .

3)  $D = \{(x, y) : x^2 + y^2 \leq 9, x \leq 0, y \leq 0\}$ .

# Zad. 1

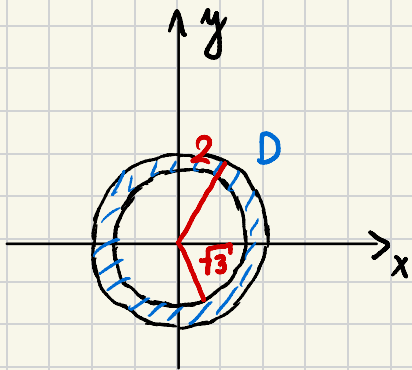
$$1) \iint_D e^{-x^2-y^2} dx dy; \quad D = \{(x,y) \mid 3 \leq x^2+y^2 \leq 4\}$$

$$x^2+y^2=3$$

$$O(0,0) \quad r=\sqrt{3}$$

$$x^2+y^2=4$$

$$O(0,0) \quad r=2$$



$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$|z| = r$$

$$0 \leq \varphi \leq 2\pi$$

$$\sqrt{3} \leq r \leq 2$$

$$\iint_D e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \left( \int_{\sqrt{3}}^2 e^{-(r^2 \cos^2 \varphi + r^2 \sin^2 \varphi)} \cdot \underbrace{r}_{\text{jacobian}} dr \right) d\varphi =$$

$$= \int_0^{2\pi} \left( \int_{\sqrt{3}}^2 e^{-r^2} \cdot r dr \right) d\varphi = \left| \begin{array}{l} t = -r^2 \\ dt = -2r dr \\ dr = \frac{dt}{-2r} \end{array} \right| \frac{r}{t} \bigg|_{-3}^{-4} =$$

$$= -\frac{1}{2} \int_0^{2\pi} \left( \int_{-3}^{-4} e^t dt \right) d\varphi = -\frac{1}{2} \int_0^{2\pi} [e^t]_{-3}^{-4} d\varphi = -\frac{1}{2} \int_0^{2\pi} (e^{-4} - e^{-3}) d\varphi =$$

$$= -\frac{1}{2} [(e^{-4} - e^{-3})\varphi]_0^{2\pi} = 11(e^{-3} - e^{-4})$$

$$2) \iint_D \sqrt{x^2+y^2} dx dy$$

$$D = \{(x,y) : x^2 + y^2 \leq 2y, x \geq 0\}$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y + 1 = 1$$

$$x^2 + (y-1)^2 = 1$$

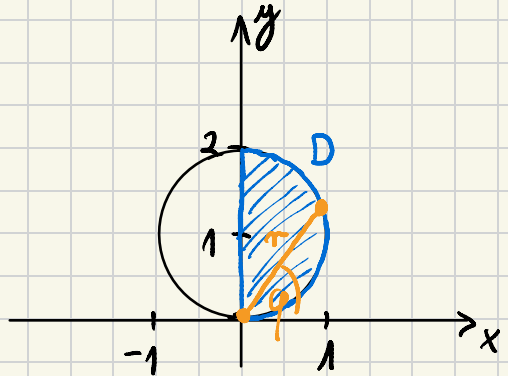
$$O(0,1) \quad r=1$$

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$|x| = r$$

$$\varphi \in [0; \frac{\pi}{2}]$$

$$r \in [0; 2 \sin \varphi]$$



$$r^2 \cos^2 \varphi + r^2 \sin^2 \varphi = 2r \sin \varphi$$

$$r^2 - 2r \sin \varphi = 0$$

$$r(r - 2 \sin \varphi) = 0$$

$$r = 0 \quad \vee \quad r = 2 \sin \varphi$$

$$\iint_D \sqrt{x^2+y^2} dx dy = \int_0^{\frac{\pi}{2}} \left( \int_0^{2 \sin \varphi} \sqrt{r^2} \cdot r dr \right) d\varphi = \int_0^{\frac{\pi}{2}} \left( \int_0^{2 \sin \varphi} r^2 dr \right) d\varphi =$$

$$= \frac{1}{3} \int_0^{\frac{\pi}{2}} [r^3]_0^{2 \sin \varphi} d\varphi = \frac{8}{3} \int_0^{\frac{\pi}{2}} \sin^3 \varphi d\varphi = \frac{8}{3} \int_0^{\frac{\pi}{2}} (1 - \cos^2 \varphi) \sin \varphi d\varphi =$$

$$= \left| \begin{array}{l} t = \cos \varphi \\ dt = -\sin \varphi d\varphi \\ d\varphi = -\frac{1}{\sin \varphi} dt \end{array} \quad \begin{array}{c|c|c} \varphi & 0 & \frac{\pi}{2} \\ \hline t & 1 & 0 \end{array} \right| = -\frac{8}{3} \int_1^0 1 - t^2 dt = -\frac{8}{3} \left[ t - \frac{1}{3} t^3 \right]_1^0 =$$

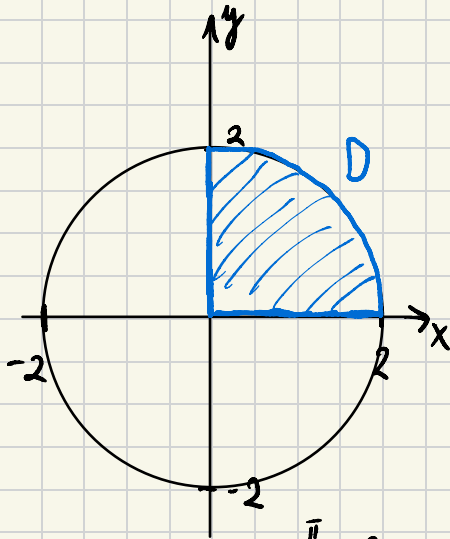
$$= -\frac{8}{3} \left( 0 - 1 + \frac{1}{3} \right) = \frac{16}{9}$$



$$3) \iint_D xy^2 dx dy$$

$$D = \{(x, y) : x^2 + y^2 \leq 4, x \geq 0, y \geq 0\}$$

$$O(0,0) \quad r=2$$



$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$|z| = r$$

$$\varphi \in [0; \frac{\pi}{2}]$$

$$r \in [0; 2]$$

$$\iint_D xy^2 dx dy = \int_0^{\frac{\pi}{2}} \left( \int_0^2 r^2 \cos \varphi \cdot r^2 \sin^2 \varphi dr \right) d\varphi =$$

$$= \int_0^{\frac{\pi}{2}} \left( \int_0^2 r^4 \sin^2 \varphi \cos \varphi dr \right) d\varphi = \frac{1}{5} \int_0^{\frac{\pi}{2}} [r^5 \sin^2 \varphi \cos \varphi]_0^2 d\varphi =$$

$$= \frac{32}{5} \int_0^{\frac{\pi}{2}} \sin^2 \varphi \cos \varphi d\varphi = \left| \begin{array}{c} t = \sin \varphi \\ dt = \cos \varphi d\varphi \\ d\varphi = \frac{1}{\cos \varphi} dt \end{array} \right| \begin{array}{c|c|c} \varphi & 0 & \frac{\pi}{2} \\ \hline t & 0 & 1 \end{array} =$$

$$= \frac{32}{5} \int_0^1 t^2 dt = \frac{32}{15} [t^3]_0^1 = \frac{32}{15}$$

$$\langle a, b \rangle \leftrightarrow (a, b) \quad \text{CIEKAWOSTKA}$$

$$[a, b] \leftrightarrow ]a, b[$$

Zad. 2

$$1) D = \{(x, y) : x + y \leq 3, y^2 \leq 4x, y \geq 0\}$$

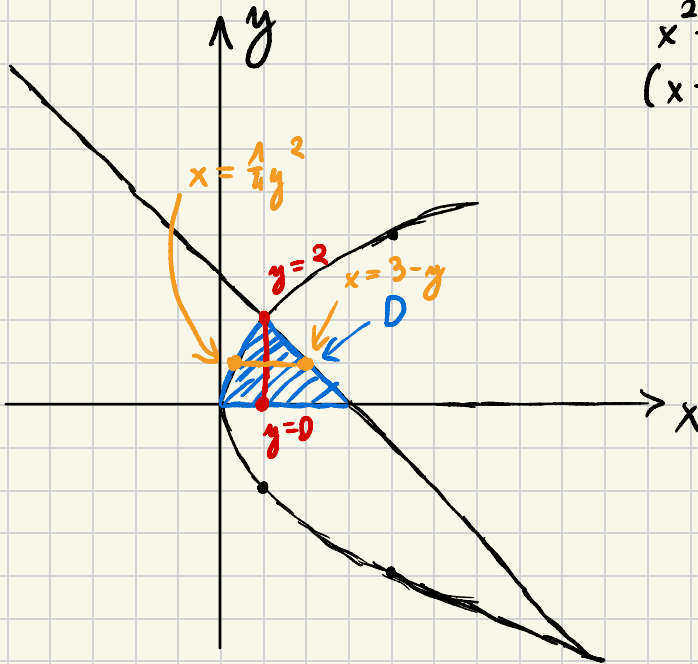
$$|D| = \iint_D 1 \, dx \, dy \quad y = 3 - x \quad x \geq \frac{1}{4}y^2$$

$$x = \frac{1}{4}(3-y)^2$$

$$4x = 9 - 6y + y^2$$

$$x^2 - 10x + 9 = 0$$

$$(x-1)(x-9) = 0$$



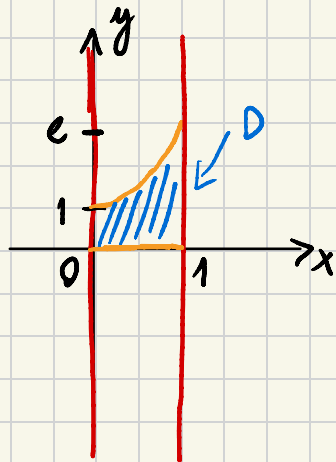
$$\begin{aligned} \iint_D 1 \, dx \, dy &= \int_0^2 \left( \int_{\frac{1}{4}y^2}^{3-y} 1 \, dx \right) dy = \int_0^2 [x]_{\frac{1}{4}y^2}^{3-y} dy = \\ &= \int_0^2 \left( 3-y - \frac{1}{4}y^2 \right) dy = \left[ 3y - \frac{1}{2}y^2 - \frac{1}{12}y^3 \right]_0^2 = \\ &= 6 - \frac{1}{2} \cdot 4 - \frac{1}{12} \cdot 8 = 6 - 2 - \frac{2}{3} = 4 - \frac{2}{3} = \frac{10}{3} \end{aligned}$$

Zad. 4

jednoduché  $\rightarrow q(x, y) = \text{const.}$

$$1) D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq e^x\}$$

$$\begin{aligned} M &= \int_0^1 \int_0^{e^x} q(x, y) dx dy = q(x, y) \int_0^1 \left( \int_0^{e^x} 1 dy \right) dx = \\ &= q(x, y) \int_0^1 [y]_0^{e^x} dx = q(x, y) \int_0^1 e^x dx = \\ &= q(x, y) [e^x]_0^1 = (e-1) q(x, y) \end{aligned}$$



$$\begin{aligned} x_0 &= \frac{1}{M} \int_0^1 \int_0^{e^x} q(x, y) x dx dy = \\ &= \frac{1}{(e-1)q(x, y)} \cdot q(x, y) \cdot \int_0^1 x dx dy = \end{aligned}$$

$$= \frac{1}{e-1} \int_0^1 \left( \int_0^{e^x} x dy \right) dx = \frac{1}{e-1} \int_0^1 [xy]_0^{e^x} dx =$$

$$= \frac{1}{e-1} \int_0^1 x \cdot e^x dx = \left| \begin{array}{l} f = x \quad f' = 1 \\ g' = e^x \quad g = e^x \end{array} \right| = \frac{1}{e-1} ([xe^x]_0^1 - \int_0^1 e^x dx) =$$

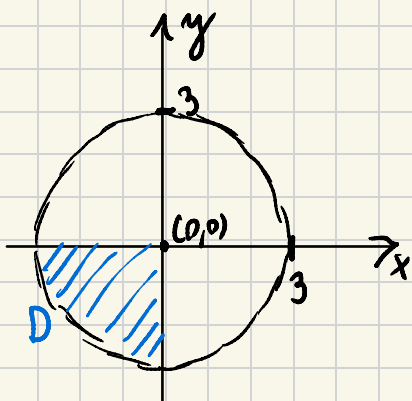
$$= \frac{1}{e-1} (e - [e^x]_0^1) = \frac{1}{e-1} (e - (e-1)) = \frac{1}{e-1}$$

$$\begin{aligned}
 y_0 &= \frac{1}{\pi} \int_0^1 \int_0^1 y \cdot q(x,y) dx dy = \frac{1}{(e-1)q(x,y)} \cdot q(x,y) \int_0^1 \int_0^1 y dx dy = \\
 &= \frac{1}{e-1} \int_0^1 \left( \int_0^1 y dx \right) dy = \frac{1}{2(e-1)} \int_0^1 [y^2]_0^1 dy = \\
 &= \frac{1}{2(e-1)} \int_0^1 e^{2x} dx = \frac{1}{4(e-1)} [e^{2x}]_0^1 = \frac{1}{4(e-1)} (e^2 - 1) = \\
 &= \frac{e+1}{4}
 \end{aligned}$$

$$(x_0, y_0) = \left( \frac{1}{e-1}, \frac{e+1}{4} \right)$$

$$3) D = \{(x,y) : x^2 + y^2 \leq 9, x \leq 0, y \leq 0\}$$

$O(0,0) \quad r=3$



$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$$

$$|z| = r$$

$$\varphi \in \left( \pi; \frac{3\pi}{2} \right)$$

$$r \in \langle 0; 3 \rangle$$

$$q = q(x,y) \text{ const.}$$

$$\begin{aligned}
 M &= \int_0^{\frac{3\sqrt{2}}{2}} \int_{\pi}^{\frac{3\sqrt{2}}{2}} g(x,y) dx dy = \int_{\pi}^{\frac{3\sqrt{2}}{2}} \left( \int_0^{\frac{3\sqrt{2}}{2}} g \cdot r dr \right) d\varphi = \\
 &= g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \left( \int_0^{\frac{3\sqrt{2}}{2}} r dr \right) d\varphi = \frac{1}{2} g \int_{\pi}^{\frac{3\sqrt{2}}{2}} [r^2]_0^{\frac{3\sqrt{2}}{2}} d\varphi = \\
 &= \frac{g}{2} \int_{\pi}^{\frac{3\sqrt{2}}{2}} d\varphi = \frac{g}{2} [\varphi]_{\pi}^{\frac{3\sqrt{2}}{2}} = \frac{g}{2} \cdot \frac{1}{2} \pi = \frac{g\pi}{4}
 \end{aligned}$$

$$\frac{1}{M} = \frac{4}{9\pi g}$$

$$x_0 = \frac{1}{M} \int_0^{\frac{3\sqrt{2}}{2}} \int_{\pi}^{\frac{3\sqrt{2}}{2}} x \cdot g(x,y) dx dy = \frac{4}{9\pi g} \cdot (9g) = -\frac{4}{\pi}$$

$$\begin{aligned}
 \int_0^{\frac{3\sqrt{2}}{2}} \int_{\pi}^{\frac{3\sqrt{2}}{2}} x \cdot g(x,y) dx dy &= g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \left( \int_0^{\frac{3\sqrt{2}}{2}} r \cos \varphi \cdot r dr \right) d\varphi = \\
 g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \left[ \frac{1}{3} \cos \varphi r^3 \right]_0^{\frac{3\sqrt{2}}{2}} d\varphi &= \frac{g}{3} \int_{\pi}^{\frac{3\sqrt{2}}{2}} 27 \cos \varphi d\varphi = \\
 = 9g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \cos \varphi d\varphi &= 9g [\sin \varphi]_{\pi}^{\frac{3\sqrt{2}}{2}} = 9g (-1 - 0) = -9g
 \end{aligned}$$

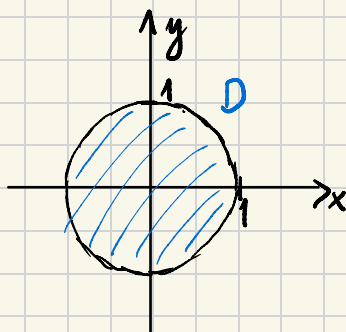
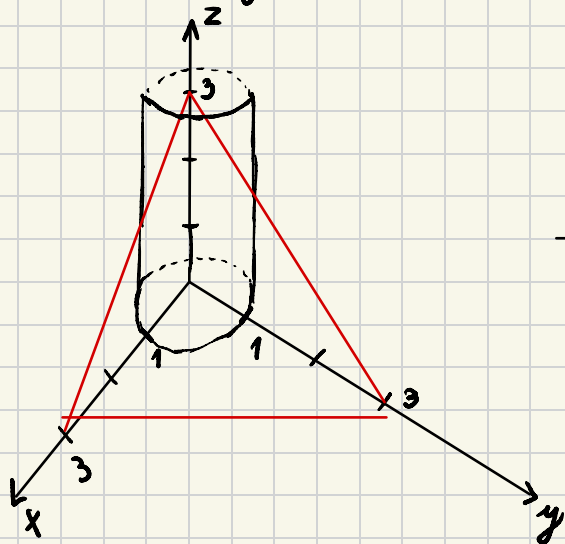
$$y_0 = \frac{1}{M} \int_0^{\frac{3\sqrt{2}}{2}} \int_{\pi}^{\frac{3\sqrt{2}}{2}} y \cdot g(x,y) dx dy = \frac{4}{9\pi g} \cdot (-9g) = -\frac{4}{\pi}$$

$$\begin{aligned}
 \int_0^{\frac{3\sqrt{2}}{2}} \int_{\pi}^{\frac{3\sqrt{2}}{2}} y \cdot g(x,y) dx dy &= g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \left( \int_0^{\frac{3\sqrt{2}}{2}} r \sin \varphi \cdot r dr \right) d\varphi = \\
 = \frac{g}{3} \int_{\pi}^{\frac{3\sqrt{2}}{2}} [r^3 \sin \varphi]_0^{\frac{3\sqrt{2}}{2}} d\varphi &= 9g \int_{\pi}^{\frac{3\sqrt{2}}{2}} \sin \varphi d\varphi = -9g [\cos \varphi]_{\pi}^{\frac{3\sqrt{2}}{2}} = \\
 &= -9g [0 + 1] = -9g
 \end{aligned}$$

### Zad. 3

1)  $x^2 + y^2 = 1$ ,  $x + y + z = 3$ ,  $z = 0$

$$\frac{x}{3} + \frac{y}{3} + \frac{z}{3} = 1$$



$$|E| = \iint_D (\text{surface} - \text{projection}) dx dy$$

surface:  $z = 3 - x - y$

projection:  $z = 0$

$$|E| = \iint_D (3 - x - y) dx dy = \int_{-1}^1 \left( \int_{-1}^1 3 - x - y dy \right) dx =$$

$$= \int_{-1}^1 \left( \left[ 3y - xy - \frac{1}{2}y^2 \right]_{-1}^1 \right) dx = \int_{-1}^1 \left( 3 - x - \frac{1}{2} + 3 - x - \frac{1}{2} \right) dx =$$

$$= \int_{-1}^1 (5 - 2x) dx = \left[ 5x - x^2 \right]_{-1}^1 = 5 - 1 + 5 - 1 = \underline{\underline{8}}$$